

# Wing Theatre Controller

## User Manual

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*Show-control software for the Behringer Wing digital mixer, built for theatre and live production.*

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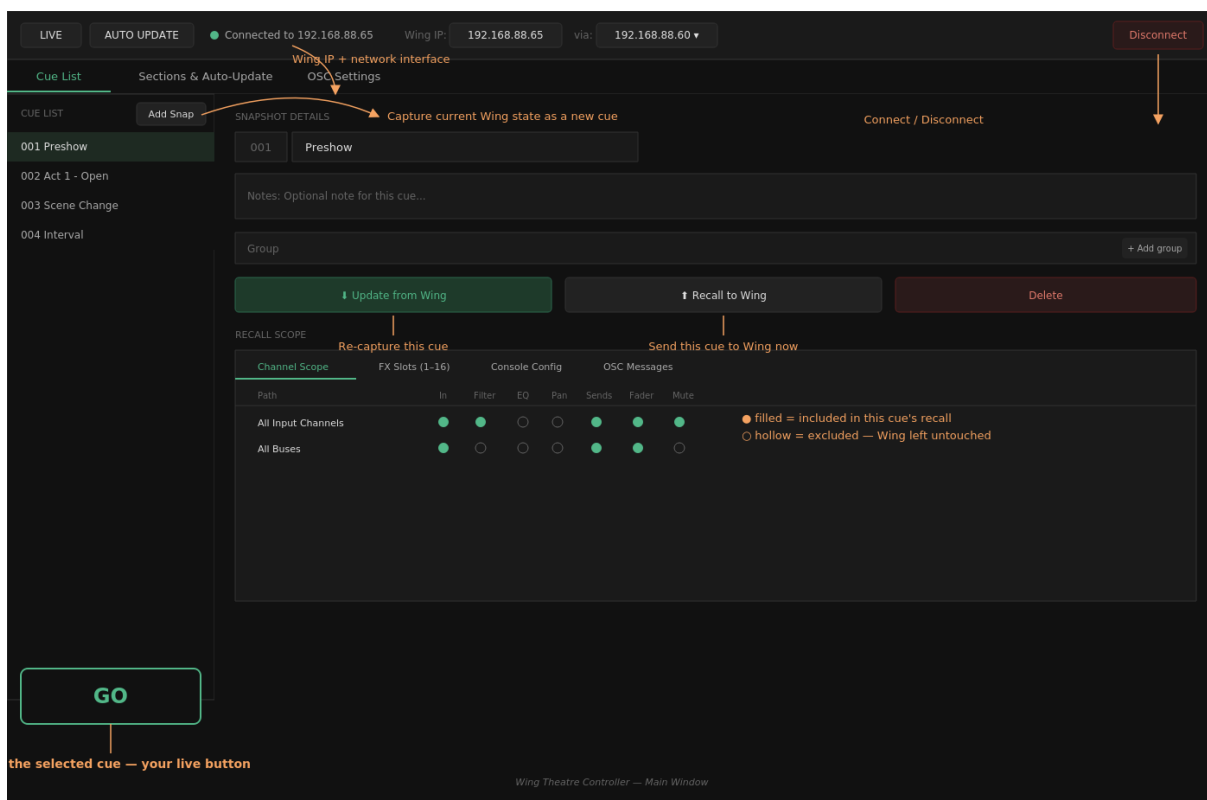
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# 1. What This Software Does

Wing Theatre Controller connects to a Behringer Wing over the network and lets you:

- Capture snapshots of the console's current state
- Build a cue list and fire snapshots in sequence with a single GO
- Choose exactly which parameters each snapshot recalls (Recall Scope) — so a cue can change only the vocal mics, only the band, only FX, etc.
- Fade faders and send levels smoothly instead of jumping
- Keep snapshots automatically up to date while you mix, using Auto Update
- Trigger cues remotely via OSC or from Bitfocus Companion (with two-way status feedback)

It runs alongside the Wing, not instead of it — you still mix on the console or in Wing Editor as normal. Wing Theatre Controller listens to everything you do and remembers it in snapshots.



Main window overview — diagrams in this manual are illustrative layouts matching the app's real colour scheme; exact spacing may differ slightly from your installed version.

## 2. Installation

### macOS

- Download WingTheatreController-macos.dmg from the release page.
- Open the DMG and drag Wing Theatre Controller into Applications.
- First launch: right-click the app → Open (unsigned app — one-time step).

## Windows

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- Download WingTheatreController-windows.zip.
- Extract it anywhere (e.g. C:\WingTheatre\).
- Run Wing Theatre Controller.exe.
- If Windows Firewall prompts you the first time you enable the Companion TCP server, click Allow access (needed so Companion on other machines can reach you).

No separate installation of Wing drivers or software is required — Wing Theatre Controller talks to the console directly over the network.

## 3. Connecting to the Wing

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At the top of the window:

| Field                | Purpose                                                                                                                                                                |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Wing IP              | The Wing console's IP address on your network                                                                                                                          |
| via                  | Which network interface on your computer to send traffic through — important if your machine has more than one active network connection (Wi-Fi + Ethernet, VPN, etc.) |
| Connect / Disconnect | Opens or closes the connection                                                                                                                                         |

Click Connect. The status dot turns green and shows Connected to [IP] once the console handshake and initial parameter sync complete (a few seconds — the app is reading the Wing's full current state).

### If it won't connect

- Confirm the Wing IP is correct and reachable (ping it)
- Check you selected the right network interface in via
- Make sure nothing else has an exclusive lock on the console connection

## 4. The Cue List

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The left-hand panel is your show's running order.

- Add Snap — captures the Wing's current state as a new cue at the end of the list
- Click a cue to select it and see its Snapshot Details on the right
- Drag to reorder cues
- Each cue has a number, an editable name, and optional Notes

### Snapshot Details panel

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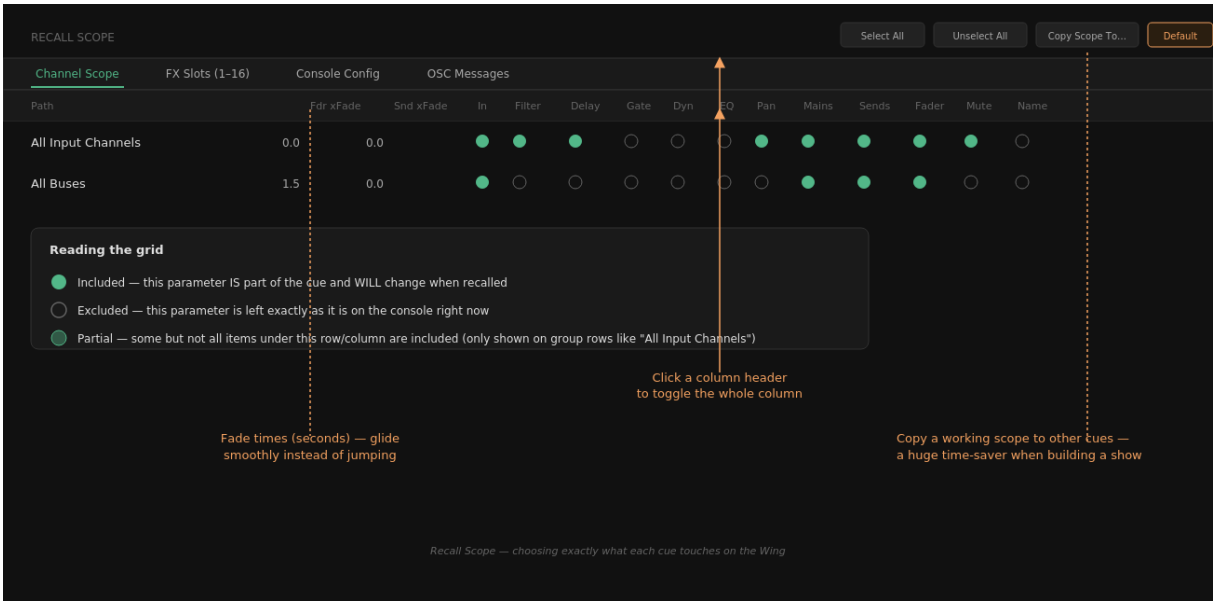
- Update from Wing — re-captures this cue from the Wing's current state (overwrites what was previously stored)
- Recall to Wing — sends this cue to the console immediately (same as selecting it and pressing GO)
- Delete — removes the cue
- Group — optionally assign the cue to a named group (see §8, Groups)

# GO

The big GO button (bottom-left) recalls the currently selected cue to the Wing. This is the button you'll use live. It respects that cue's Recall Scope and any fade times configured for it.

## 5. Recall Scope — Choosing What Each Cue Changes

This is the heart of the software. Every cue has its own Recall Scope: a per-parameter map of what gets sent to the Wing and what's left untouched.



The Recall Scope grid

Four tabs under Recall Scope:

| Tab             | Covers                                                                                                                                                                   |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Channel Scope   | Channels, buses, matrix outputs, main outputs, DCA groups — broken down by function (In, Filter, Delay, Gate, Dyn, EQ, Pan, Mains, Sends, Fader, Mute, Name, Tags, etc.) |
| FX Slots (1-16) | The 16 internal FX processors                                                                                                                                            |
| Console Config  | Global console settings                                                                                                                                                  |
| OSC Messages    | Custom OSC messages sent to external gear when this cue recalls                                                                                                          |

Each cell is a toggle:

- Included (solid green) — this parameter is part of the cue and will be recalled
- Excluded (hollow) — this parameter is ignored; the console stays as it is
- ◐ Partial — some but not all items in that row/column are included

Click a column header (e.g. "EQ") to toggle that entire column at once. Click the small expand arrow next to a row (e.g. "All Input Channels") to expand it to individual channels.

### Toolbar buttons

- Select All / Unselect All — set every cell in the current tab

- Copy Scope To... — copy this cue's scope to other cues (huge time-saver for building a show)
- Default — left-click resets this cue's scope to a 1-to-1 copy of the app's current default scope

### Setting your own default scope

Right-click the Default button to open "Edit Default Recall Scope." This is a full scope editor — the same grid as on a normal cue — where you set what the default should be, including per-group differences (e.g. Sends off for buses only) and default fade times. FX Slots and Console Config have their own defaults too; OSC Messages are never part of the default.

This matters beyond just the Default button: every new cue — including ones created with Add Snap — starts out with whatever the current default scope is. Set it once to match how you like to work, and every cue you create from then on already has the right scope, instead of you configuring it by hand every time.

Changes made in this dialog only take effect on cues where you explicitly press Default (or on new cues going forward) — existing cues you've already configured are never changed automatically.

### Why this matters

A cue with only Mute and Fader included on "All Input Channels" will only move faders and mutes when recalled — EQ, dynamics, pan, everything else on the console is left completely alone. This lets you build cues that touch just what a scene actually needs.

## 6. Fades

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Two fade-time columns appear per row in Channel Scope:

- Fdr xFade — fade time (seconds) for that channel/bus's fader level
- Snd xFade — fade time (seconds) for its send levels

Set a fade time greater than 0.0 and that parameter will glide smoothly to its target value over that many seconds when the cue recalls, instead of jumping instantly. Fades interpolate linearly in dB, so they sound and look natural.

### Fade + Mute coordination

If a cue simultaneously fades a fader and changes its mute state, the software times the mute intelligently:

- Fading down into a mute: the mute is applied right at the end of the fade — so you hear the level drop away before the channel goes silent, not an abrupt cut mid-fade.
- Fading up out of a mute: the channel is unmuted immediately, so the fade-in is actually audible from the start rather than silent while the fader ramps.

Pressing GO on a new cue automatically cancels any fades still in progress from the previous cue.

## 7. Auto Update

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Found under the Sections & Auto-Update tab.

Auto Update keeps your already-recalled cues in sync with live changes you make on the console — so if you rode a fader during a scene, the next time that cue is recalled it remembers where you left it, rather than snapping back to the originally-saved value.

- Toggle AUTO UPDATE on/off from the top toolbar or remotely via OSC/TCP

## Sections are required

Auto Update will not write anything at all until at least one Section exists in the show. With no sections defined, the status bar shows "create a section first to enable per-cue writing" and every live change is ignored. Create a Section under the Sections & Auto-Update tab to turn writing on.

## What a Section controls

Each Section sets, per parameter type (fader, mute, sends, EQ, dynamics, etc.), how far a live change is written:

- Snap — only the currently active (last GO'd) cue is updated
- Group — every cue sharing the same Group as the active cue is updated (see §8, Groups)
- All — every cue in the entire show is updated

By default, Fader, Mute, Sends and FX Rack are set to Snap, and everything else defaults to All — a sensible starting point where level rides stay local to the current cue, while broader setup changes propagate everywhere. Adjust this per Section to match how a scene actually behaves. A Section can also be restricted to specific channels only.

## A cue must be active

Auto Update only ever writes to cues once GO has been pressed at least once — moving a fader before any cue is active does nothing. If you see "press GO on a cue first to enable writing" in the status bar, that's why.

# 8. Groups

Cues can be assigned to a named Group (set in the Snapshot Details panel). Groups are primarily an organisational tool — useful for filtering, batch scope operations, and visually grouping related cues (e.g. all cues belonging to one song or scene) in a large show.

# 9. OSC Settings

OSC OUTPUT ENDPOINTS

(targets for custom OSC Messages, see §9)

**REMOTE CONTROL — incoming OSC to control Wing Theatre**

Listen port: 9000

Listening on 0.0.0.0:9000

```
/wingtheatre/go -- fire GO
/wingtheatre/next/go -- play next cue
/wingtheatre/snap/go <name> -- recall specific cue
/wingtheatre/autoupdate/on -- toggle Auto Update
/wingtheatre/addsnap [name] -- add new snapshot
```

One-way triggers — great for a lighting desk or stage-manager panel sending simple cues

**COMPANION / TCP REMOTE CONTROL — single client with feedback**

Listen port: 9001

Connected: 192.168.88.60

Commands (TCP, newline-delimited):

```
GO / NEXT_GO / PREV_GO / SNAP_GO <name>
AU_ON / AU_OFF / AU_TOGGLE / ADD_SNAP [name]
GET_STATE
```

Feedback sent on every change:

```
STATE current_cue_num=001
STATE current_cue_name=Preshow
STATE wing_connected=true
STATE autoupdate=false
```

Two-way — Companion buttons show live status back (current cue, connection, Auto Update)

**Setting up in Bitfocus Companion**

1. Companion → Connections → Add "Wing Theatre Controller"
2. IP = the computer running Wing Theatre Controller. Port = the TCP port above (default 9001)
3. Enable the connection — buttons with feedback are now available in Companion's button designer

OSC Settings — remote triggering via UDP (OSC) or TCP (Companion)

OSC and TCP remote control settings

## OSC Output Endpoints

External OSC targets (other consoles, media servers, lighting desks) that receive messages when a cue recalls, if that cue has custom messages defined in its OSC Messages scope tab.

- + Add Endpoint — add a target (Name, IP Address, Port, Via interface)
- Toggle Enabled per endpoint
- Remove Selected to delete

## Remote Control (UDP) — incoming OSC

Lets an external device (lighting console, stage manager panel, etc.) drive Wing Theatre Controller over OSC.

- Listen port — UDP port to listen on (default 9000)
- Toggle On/Off

| OSC Address                    | Action                                     |
|--------------------------------|--------------------------------------------|
| /wingtheatre/go                | Fire GO (play current/selected cue)        |
| /wingtheatre/previous/go       | Play the cue before the last-fired one     |
| /wingtheatre/next/go           | Play the cue after the last-fired one      |
| /wingtheatre/previous          | Move selection up (no recall)              |
| /wingtheatre/next              | Move selection down (no recall)            |
| /wingtheatre/snap/go <n name>  | Recall a specific cue by number or name    |
| /wingtheatre/store             | Store current Wing state as a new snapshot |
| /wingtheatre/autoupdate/on off | Toggle Auto Update                         |
| /wingtheatre/addsnap [name]    | Add a new snapshot (optional name)         |

## Companion / TCP Remote Control — incoming TCP

A dedicated TCP protocol used by the Bitfocus Companion module, offering two-way state feedback (not just one-way triggering like OSC).

- Listen port — TCP port to listen on (default 9001)
- Toggle On/Off

| Command          | Action                          |
|------------------|---------------------------------|
| GO               | Fire the currently selected cue |
| NEXT_GO          | Fire the next cue               |
| PREV_GO          | Fire the previous cue           |
| SNAP_GO <n name> | Fire a specific cue             |



|                            |                                 |
|----------------------------|---------------------------------|
| NEXT                       | Move selection down (no recall) |
| PREV                       | Move selection up (no recall)   |
| AU_ON / AU_OFF / AU_TOGGLE | Control Auto Update             |
| ADD_SNAP [name]            | Add a snapshot                  |
| GET_STATE                  | Request a full state refresh    |

Feedback sent automatically on every change: current\_cue\_num, current\_cue\_name, next\_cue\_num, next\_cue\_name, selected\_cue\_num, selected\_cue\_name, autoupdate, wing\_connected, cue\_count, fading

## 10. Bitfocus Companion Integration

A dedicated Companion module (v5+) ships alongside the app, giving you buttons with live feedback — the button text/colour updates automatically to reflect the current cue, connection state, and Auto Update status.

### Installing the module

Companion loads third-party modules from a folder on disk — the module isn't installed through Companion's own interface, so this is a one-time manual setup per machine running Companion:

- Download companion-module-wingtheatre.zip from the Releases page and extract it anywhere (e.g. a companion-modules folder)
- Install the module's dependencies using Companion's own bundled Node.js — it doesn't need a separate Node/Yarn install:
  - Windows: double-click install-windows.bat inside the extracted folder — it finds Companion automatically and installs everything
  - macOS: open Terminal in the extracted folder and run the two commands below

```
/Applications/Companion.app/Contents/Resources/node-runtimes/node22/bin/node \
/Applications/Companion.app/Contents/Resources/node-runtimes/node22/lib/
node_modules/corepack/dist/corepack.js \
yarn install
```

- Open Companion Launcher (not the browser UI) → the gear/cog icon → Advanced Settings
- Under Developer, click Select and choose the PARENT folder that contains the extracted module folder (not the module folder itself)
- Enable Developer Modules, then click Launch GUI

The module now appears as an available connection type in Companion, and stays available across restarts — this only needs doing once per machine, and again after installing a newer version of the module.

### Adding the connection

- In Companion → Connections, add Wing Theatre Controller
- Set the IP address to the computer running Wing Theatre Controller, and the port to the TCP port set in OSC Settings (default 9001)
- Enable the connection

### Available Actions

GO, Next GO, Previous GO, Snap GO, Auto Update ON/OFF/Toggle, Next cue (no GO), Previous cue (no GO), Add Snap, Get State

## Available Variables

`$(wingtheatre:current_cue_num)`, `current_cue_name`, `next_cue_num`, `next_cue_name`, `selected_cue_num`, `selected_cue_name`, `autoupdate`, `wing_connected`, `cue_count`, `fading`

## Available Feedbacks

`autoupdate`, `wing_connected`, `fading`, `current_cue_num`, `current_cue_name` — use these to colour-code buttons (e.g. red when Wing is disconnected, green when Auto Update is on).

*Note: Companion talks to Wing Theatre Controller, not directly to the Wing console — Wing Theatre Controller is always the single source of truth in the middle.*

# 11. Show Files

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- New Show — start a blank cue list
- Open / Save / Save As — standard show-file management (.wts files)
- Autosave — set an interval (minutes) for automatic background saves; 0 min disables it

Show files store the full cue list, every cue's snapshot data, its recall scope, fade times, groups, sections, and OSC message configuration.

# 12. About / Version

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Click the ? button in the toolbar to see the installed version number and build date, plus credits and a link to the project's GitHub page. Useful when reporting an issue — always include this version string.

# 13. Troubleshooting

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| Symptom                                            | Likely cause / fix                                                                                                |
|----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Won't connect to Wing                              | Wrong IP, wrong network interface selected in via, or Wing unreachable on the network                             |
| GO does nothing                                    | Check the connection status dot is green; check the cue's Recall Scope isn't fully excluded                       |
| Companion shows no connection                      | Confirm the TCP port matches on both ends; check firewall allowed the app through                                 |
| A parameter recalls to the wrong value             | Confirm it's actually included in that cue's Recall Scope — an excluded parameter is always left as-is            |
| App disconnects after a few minutes of no activity | Should not happen on current versions — if it does, please report with the version number from the About dialog   |
| EQ/Dynamics values look wrong after recall         | Confirmed fixed as of the current build — please report if still seen, along with the exact parameter and version |

# 14. Credits

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- Wing Theatre Controller — Mikkel Peter Larsen & Claude (Anthropic)
- wingmon (Wing protocol bridge) — Mikkel Peter Larsen & Claude, built on libwing by dannyfiresnake

- Protocol documentation — Patrick-Gilles Maillot
- python-osc — attwad
- PyQt6 — Riverbank Computing

Source and releases: [github.com/mikk3722/wing-theatre-controller](https://github.com/mikk3722/wing-theatre-controller)